



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

BIG ORANGE. BIG IDEAS.

COSC 522, Fall 2026
University of Tennessee, Knoxville

Meeting Time and Place: MKB-404; Tue/Thur 11:20 AM - 12:35 PM

Course Credit Hours: 3.0

Faculty Contact Information

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TA Office Hours:

Sanskar Amgain: TBD

Nan Xiao: TBD

Course Description/Information:

Machine learning is the study of adaptive algorithms that automatically improve with experience. This field spans a wide array of technologies, from classical statistics problems to complex

robotics challenges and modern frontiers in personalized medicine, materials discovery, and generative media.

This course offers a rigorous foundation in the methods, mathematics, and algorithms of modern machine learning. It is **intended for graduate students (both MS and PhD)**, as well as **ambitious undergraduate students** with the necessary background. While a working knowledge of probability, statistics, and algorithms is an advantage, the class is structured so that anyone with a good mathematical and computer science foundation can fully participate.

Prerequisites

To succeed in this course, you should meet the following prerequisites:

- **Strong Programming Skills:** Proven ability to write non-trivial programs. Coursework and assignments will require proficiency in **Python**.
- **Mathematical Maturity:** Good knowledge about Probability & Statistics, Linear Algebra, Calculus and Discrete Mathematics.

Student Learning Outcomes/Objectives:

By the end of the course, students should be able to:

- Understand how popular machine learning algorithms work, what they assume, and when to use them.
- Implement and test key algorithms for classification, regression, and clustering.
- Build a complete machine learning pipeline, from cleaning data to choosing the best model.
- Use math (like probability, calculus, and linear algebra) to analyze and create new models.
- Explain the difference between major learning types, such as supervised and unsupervised.
- Design experiments to properly test and compare different machine learning methods.
- Read and analyze current research papers in the field to understand the latest trends.

Course Communications:

This term we will be using [Discord](#) for class discussion. This system is highly catered to getting you help fast and efficiently from classmates and myself. Rather than emailing questions to the instructor, I encourage you to post your questions on Discord. Find our class signup link [here](#). The instructor will also be available during office hours.

Students are also free to email me directly for other matters. You can expect a response in about one business day. If a more immediate response is needed please come by my office. For technical issues, contact the OIT HelpDesk via phone (865) 974-9900 or online at <http://help.utk.edu/>.

How to Be Successful in This Course:

Attend the lectures and ask any questions you had about the material. For projects and homeworks, start early. This will enable you to get the help you need. Most importantly, if you need help, come and ask for it. My door is always open and I welcome students to come discuss struggles they are having or just suggestions you may have for the class.

In general, the University's common responsibilities apply:

Student's Responsibility

- Be prepared for all classes
- Be respectful of others
- Actively contribute to the learning activities in class
- Abide by the UT Honor Code

Instructor's Responsibility

- Be prepared for all classes
- Evaluate all fairly and equally
- Be respectful of all students
- Create and facilitate meaningful learning activities
- Behave according to University codes of conduct

Texts/Resources/Materials:

This course does not exactly follow any one textbook. However, most lectures will have some optional reading to help you better understand the material or see a different presentation/perspective. Many of the textbooks are freely available online:

- [Pattern Recognition and Machine Learning](#); Christopher M. Bishop.
- [A Course in Machine Learning](#); Hal Daumé III.
- [Machine Learning](#); Tom Mitchell.
- [Machine Learning: a Probabilistic Perspective](#); Kevin P. Murphy.
- [Dive into Deep Learning](#); Aston Zhang, Zack C. Lipton, Mu Li, Alex J. Smola.

The textbook below is a great resource for those hoping to brush up on the prerequisite mathematics background for this course:

- [Mathematics for Machine Learning](#), Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

However, if you find yourself regularly and frequently referring to this textbook, we strongly encourage you to consider whether or not you have the necessary prerequisite knowledge to succeed in this course.

Course Requirements, Assessments, and Evaluations:

- Midterm = 30%
- Final Project = 30%
- 3 Homework Assignments = 30%
- In class quizzes (5% for attendance, 5% for correctness) = 10%

Homework Assignments:

Each homework assignment will consist of coding tasks and/or theoretical derivations.

Homework Late Policy: Please refer to the class schedule on the due dates for homeworks.

- Students can submit late with the penalty of 25% deduction for every 24 hours late (up to 3 days).
- After 3 days, no more late submission is allowed.
- Extensions can be granted for special cases (email the instructor)

Projects:

The final project is your opportunity to tackle a challenging machine learning problem in depth, culminating in a conference-style paper and a documented code repository. Students are strongly encouraged to propose their own unique topics, or may choose from a provided list. Cumulatively, the final project will account for **30%** of a student's final grade. Each group should consist of a max of **5** students. Related team information should be entered [here](#).

All projects fall into one of two tracks:

1. **Track 1: Methodological Advancement:** This track focuses on advancing ML science. Projects involve a deep investigation of algorithms, such as proposing a novel technique, rigorously reproducing and extending an influential paper, or conducting a large-scale empirical study to compare existing methods. The contribution is a new insight or improvement to ML methods themselves.
2. **Track 2: Novel Application:** This track focuses on advancing ML practice. Projects involve building a full-pipeline solution to a real-world problem, requiring data collection, cleaning, and modeling. Examples include analyzing scientific data from materials science or building a predictive system for a local Knoxville challenge.

Extra point opportunities:

1. +5%: final project leads to submission to a workshop (**requires instructor approval**)
2. +10%: final project leads to submission to a top-tier conference (**requires instructor approval**)

Letter grade distribution: student scores will be converted into the letter grade based on the following criteria:

- A [90%, 100%]
- A- [85%, 90%)

The instructor reserves the right to revise, alter, or amend this syllabus as necessary.

- B+ [80%, 85%)
- B [75%, 80%)
- B- [70%, 75%)
- C+ [65%, 70%)
- C [60%, 65%)
- D [55%, 60%)
- F [0%, 55%)

Course Feedback:

At any point in the course, feel free to come talk to me or email me with your feedback.

Course Schedule:

See Canvas for the course schedule. This schedule is fluid, changing as needed to accommodate student needs.

University Policies:

Academic Integrity:

An essential feature of the University of Tennessee, Knoxville is a commitment to maintaining an atmosphere of intellectual integrity and academic honesty. As a student of the university, I pledge that I will neither knowingly give nor receive any inappropriate assistance in academic work, thus affirming my own personal commitment to honor and integrity.

University Civility Statement:

Civility is genuine respect and regard for others: politeness, consideration, tact, good manners, graciousness, cordiality, affability, amiability and courteousness. Civility enhances academic freedom and integrity and is a prerequisite to the free exchange of ideas and knowledge in the learning community. Our community consists of students, faculty, staff, alumni, and campus visitors. Community members affect each other's well-being and have a shared interest in creating and sustaining an environment where all community members and their points of view are valued and respected. Affirming the value of each member of the university community, the campus asks that all its members adhere to the principles of civility and community adopted by the campus:

<http://civility.utk.edu/>.

Disability Services:

The University of Tennessee, Knoxville, is committed to providing an inclusive learning environment for all students. If you anticipate or experience a barrier in this course due to a chronic health condition, a learning, hearing, neurological, mental health, vision, physical, or other kind of disability, or a temporary injury, you are encouraged to contact Student Disability Services (SDS) at 865-974-6087 or sds@utk.edu. An SDS Coordinator will meet with you to develop a plan to ensure you have equitable access to this course. If you are already registered with SDS, please contact your instructor to discuss implementing accommodations included in your course access letter.

Your Role in Improving Teaching and Learning Through Course Assessment:

At UT, it is our collective responsibility to improve the state of teaching and learning. During the semester, you may be requested to assess aspects of this course either during class or at the completion of the class. You are encouraged to respond to these various forms of assessment as a means of continuing to improve the quality of the UT learning experience.

Key Campus Resources for Students:

- [Center for Career Development](#) (Career counseling and resources; HIRE-A-VOL job search system)
- [Course Catalogs](#) (Listing of academic programs, courses, and policies)
- [Hilltopics](#) (Campus and academic policies, procedures and standards of conduct)
- [OIT HelpDesk](#) (865) 974-9900
- [Schedule of Classes/Timetable](#)
- [Student Health Center](#) (visit the site for a list of services)
- [Student Success Center](#) (Academic support resources)
- [Undergraduate Academic Advising](#) (Advising resources, course requirements, and major guides)
- [University Libraries](#) (Access to library resources, databases, course reserves, and services)